LLD 3.4 (generated)

Generate LLD3.4 based on LLD3.1,LLD3.2and LLD3.3

DOC 3.4 — Low Level Design: Alerting & Monitoring Service

1. Scope

Detailed design of the alerting and monitoring service of the ADMP: the Alerting & Monitoring Service that evaluates configured parameter thresholds against ingested data and raises alerts on exceedance. Realises FR-8, IF-5 and acceptance criterion AC-3. Conforms to the HLD “Alerting & Monitoring” component.

2. Module structure

- `alerting/evaluator.py` — evaluates threshold breaches against historical data (FR-8).
- `alerting/notifier.py` — sends alerts via email, webhook or message queue (IF-5).
- `alerting/repository.py` — transactional alert writes to the catalog.
- `catalog/models.py` — `Alert`, `ThresholdDefinition`, `ParameterExceedance`.
- `catalog/lineage.py` — links each alert to its raw source object.

3. Interfaces

- `evaluate_alerts(alert_config: dict, records: list[ADMRecord]) -> AlertList`.
- `send_notification(notification: Notification) -> None`.
- `upsert_alert(alert: Alert) -> None`.

4. Data model

Catalog (relational):

- `alert(alert_id PK, flight_id FK, parameter_id FK, threshold_definition_id FK, ts_range, raw_ref)`
- `threshold_definition(threshold_id PK, name, unit, source, min, max)`
- `parameter_exceedance(parameter_id FK, alert_id FK, ts_range, value, source_object (lineage))`

5. Sequence

1. Evaluator receives a batch of records from the broker and evaluates each against configured thresholds.
2. On breach, Notifier sends an alert via email/webhook/message queue to the relevant recipient(s).
3. Repository upserts the alert into the catalog with lineage back to its raw source.

```
[broker] -> evaluator -> notifier (email/webhook/queue) ->
[recipient]
                         |
                         +-> repository -> catalog (alert,
threshold_definition, parameter_exceedance)
```

6. Non-functional design notes

- **Performance (NFR-1 support):** uses a sliding window of historical data to bound evaluation cost.
- **Security (NFR-3):** alerts are sent via secure channels; recipient credentials are managed by the corporate identity provider.
- **Observability (NFR-5):** each alert carries a correlation id from ingestion onward.

7. Requirements traceability

Requirement	How addressed
FR-8	Evaluator evaluates threshold breaches against historical data.
IF-5	Notifier sends alerts via email/ webhook/message queue.
AC-3	Threshold breach raises an alert within 60 seconds of ingestion.

Requirements coverage

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FR-8	Evaluator evaluates threshold breaches against historical data.
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AC-3	Threshold breach raises an alert within 60 seconds of ingestion.

Assumptions

- The Alerting & Monitoring Service is deployed symmetrically across two sites.
- The catalog store uses point-in-time backups with daily fulls and continuous transaction-log backup, retained 35 days.
- The evaluator uses a sliding window of historical data to bound evaluation cost.
- The notifier sends alerts via secure channels; recipient credentials are managed by the corporate identity provider.